G-Code Viewer

User Manual

Black Sphere Industries

Version 1.0 — March 2026

Table of Contents

1 Introduction . . . 4
1.1 Key Capabilities . . . 4
2 Getting Started . . . 4
2.1 Accessing the Application . . . 4
2.2 Loading G-Code . . . 4
2.3 System Requirements . . . 4
3 Features . . . 4
3.1 G-Code Parser . . . 4
3.1.1 G-Codes (Preparatory) . . . 5
3.1.2 M-Codes (Machine) . . . 5
3.1.3 Address Codes . . . 5
3.2 Visualization . . . 5
3.2.1 2D View . . . 5
3.2.2 3D View . . . 6
3.3 Simulation Engine . . . 6
3.3.1 Playback Controls . . . 6
3.3.2 Speed Control . . . 6
3.3.3 Progress Indicator . . . 6
3.4 Real-Time State Display . . . 6
3.5 Statistics . . . 6
3.6 Tool Settings . . . 7
3.7 Syntax Highlighting . . . 7
3.8 Export . . . 7
3.9 Built-In Examples . . . 7
4 User Interface . . . 7
4.1 Layout . . . 7
4.2 Header . . . 7
4.3 Theming . . . 7
5 Workflows . . . 8
5.1 Previewing a CNC Program . . . 8
5.2 Step-by-Step Simulation . . . 8
5.3 Verifying Feed Rates . . . 8
5.4 Checking Bounding Box . . . 8
6 Architecture . . . 8
6.1 Architecture Overview . . . 9
6.2 Sequence Diagram — Main Parse and Render Flow . . . 10
6.3 Sequence Diagram — File Upload . . . 12
6.4 Sequence Diagram — Simulation . . . 13
6.5 State Diagram . . . 15
7 Configuration . . . 15
7.1 Supported File Extensions . . . 15
7.2 Default Tool Settings . . . 16
8 Troubleshooting . . . 16
8.1 G-Code Not Parsing . . . 16
8.2 3D View Not Rendering . . . 16
8.3 Simulation Stuttering . . . 16

8.4 Incorrect Bounding Box 16
8.5 File Upload Not Working 16

1 Introduction

G-Code Viewer is a Progressive Web App (PWA) developed by Black Sphere Industries for previewing and simulating CNC toolpaths. It parses standard G-code files and renders the resulting toolpaths in both 2D and 3D views, with a full-featured simulation engine for step-by-step or continuous playback.

The application supports the most common G-code commands used in CNC milling and 3D printing, including linear and arc interpolation, tool changes, coordinate systems, and work offsets.

1.1 Key Capabilities

- Parse and validate G-code from text input or file upload
- 2D top-down toolpath visualization with color-coded moves
- 3D perspective toolpath visualization
- Simulation playback with play, pause, stop, step forward, and step back controls
- Adjustable simulation speed (0.01x to 100x)
- Real-time machine state display (X, Y, Z, F, S, T)
- Comprehensive statistics: total distance, cut distance, rapid distance, bounding box, machine time, command count, tool changes, feed rate range
- Tool diameter configuration
- Syntax-highlighted G-code display
- Three built-in example programs
- Export functionality
- File drag-and-drop upload (.nc, .gcode, .tap, .ngc)
- Installable as a PWA
- Dark theme with amber accents

2 Getting Started

2.1 Accessing the Application

G-Code Viewer is accessible at its standalone URL or through the BSI Portal app grid. The application runs entirely in the browser with no server-side processing.

2.2 Loading G-Code

There are three ways to load G-code into the viewer:

1. **Paste Tab** — Paste G-code text directly into the editor textarea, then click "Parse & Render."
2. **Upload Tab** — Click the upload area or drag and drop a G-code file. Supported formats: `.nc`, `.gcode`, `.tap`, `.ngc`, `.txt`.
3. **Examples** — Click the "Examples" button in the header to load a built-in example program.

2.3 System Requirements

A modern browser with Canvas API support is required. Three.js is used for 3D rendering. No plugins or extensions are needed.

3 Features

3.1 G-Code Parser

The built-in parser processes standard G-code line by line, handling:

3.1.1 G-Codes (Preparatory)

Code	Function
G0	Rapid positioning (traverse)
G1	Linear interpolation (cutting move)
G2	Clockwise arc interpolation
G3	Counter-clockwise arc interpolation
G17	XY plane selection
G18	XZ plane selection
G19	YZ plane selection
G20	Inch units
G21	Millimeter units
G28	Return to home position
G54–G59	Work coordinate system selection
G90	Absolute positioning mode
G91	Incremental positioning mode
G92	Set position (coordinate offset)

3.1.2 M-Codes (Machine)

Code	Function
M0/M1	Program stop / optional stop
M2/M30	Program end
M3	Spindle on (clockwise)
M4	Spindle on (counter-clockwise)
M5	Spindle off
M6	Tool change
M8	Coolant on
M9	Coolant off

3.1.3 Address Codes

The parser recognizes X, Y, Z, A, B, C axis positions, I, J, K arc center offsets, F feed rate, S spindle speed, T tool number, and P/Q dwell and cycle parameters.

Comments are stripped (text after `;` or within parentheses).

3.2 Visualization

3.2.1 2D View

The 2D view renders a top-down (XY plane) projection of the toolpath on a Canvas element. Features include:

- Color-coded paths: rapid moves vs. cutting moves.
- Current position indicator during simulation.
- Pan by click-drag and zoom by scroll wheel.
- Tool diameter visualization.

3.2.2 3D View

The 3D view renders the full XYZ toolpath with:

- Perspective projection with orbit controls.
- Color-coded path segments.
- Axis indicators.
- Interactive rotation, zoom, and pan.

Switch between views using the "2D View" and "3D View" tabs.

3.3 Simulation Engine

The simulation engine provides frame-by-frame or continuous playback of the parsed toolpath.

3.3.1 Playback Controls

- **Play** — Start continuous simulation from the current position.
- **Pause** — Pause the simulation at the current line.
- **Stop** — Reset the simulation to the beginning.
- **Step Back** — Move one G-code command backward.
- **Step Forward** — Advance one G-code command forward.

3.3.2 Speed Control

The speed slider adjusts playback rate from 0.01x (very slow, for detailed inspection) to 100x (fast overview). The current speed is displayed next to the slider.

3.3.3 Progress Indicator

A progress bar and line counter (e.g., "47/230") show the current position within the G-code program.

3.4 Real-Time State Display

The "Current State" panel shows live machine state during simulation:

- **X, Y, Z** — Current tool position coordinates.
- **F** — Active feed rate.
- **S** — Spindle speed.
- **T** — Active tool number.

Values update in real time as the simulation advances.

3.5 Statistics

After parsing, the statistics panel displays:

Total Distance Combined length of all tool movements in mm.
Cut Distance Distance of cutting moves (G1, G2, G3) in mm.
Rapid Distance Distance of rapid traverse moves (G0) in mm.
Bounding Box The X, Y, and Z extents of the toolpath in mm.
Machine Time Estimated machining time based on feed rates in seconds.
Commands Total number of G-code lines and count of valid commands.
Tool Changes Number of M6 tool change commands.
Feed Rate Range Minimum and maximum feed rates used in the program.

3.6 Tool Settings

The tool diameter input (default: 3.175 mm) affects the toolpath width rendering in the visualization. Adjust this to match your actual tool for more accurate visual representation.

3.7 Syntax Highlighting

The G-code editor displays parsed code with color-coded syntax:

- G-codes in amber
- M-codes in green
- Coordinates in cyan
- Feed rates in yellow
- Comments in gray

The current simulation line is highlighted with an amber border.

3.8 Export

The Export button in the header allows exporting the visualization or parsed data for use in other applications.

3.9 Built-In Examples

Three example programs are included for quick evaluation:

Rectangle Pocket A simple rectangular pocket milled to depth. Demonstrates linear interpolation and depth stepping.

Circle & Drilling A circular pocket with a drilling pattern. Shows arc interpolation and canned cycles.

Contour with Arcs A complex contour using arc interpolation (G2/G3). Demonstrates clockwise and counter-clockwise arcs.

4 User Interface

4.1 Layout

The application uses a three-panel layout:

- **Left panel** (1/3 width) — G-code editor with Paste and Upload tabs.
- **Center panel** (flexible) — Visualization canvas with 2D/3D view tabs.
- **Right panel** (320px) — Simulation controls, tool settings, machine state, and statistics.

4.2 Header

The header bar contains:

- Terminal-style dots (red, yellow, green) for visual identity.
- Application title “G-CODE VIEWER” in amber monospace.
- Export button.
- Examples button for loading built-in programs.

4.3 Theming

The application uses a dark theme with:

- Background: zinc-950 (#09090b)
- Accent: amber-500 (#f59e0b)
- Text: zinc-100 (#e4e4e7)

- Code highlighting with distinct colors for G-codes, M-codes, coordinates, and comments.

5 Workflows

5.1 Previewing a CNC Program

1. Open G-Code Viewer.
2. Click the "Upload File" tab.
3. Drag and drop your `.nc` or `.gcode` file onto the drop zone.
4. The G-code is parsed automatically and the toolpath is rendered.
5. Switch between 2D and 3D views to inspect the toolpath.
6. Review statistics for distance, bounding box, and estimated time.

5.2 Step-by-Step Simulation

1. Load a G-code program (paste, upload, or example).
2. Click **Step Forward** to advance one command at a time.
3. Observe the current state panel updating with X, Y, Z positions.
4. The current line is highlighted in the G-code editor.
5. Click **Step Back** to go back and re-examine a move.
6. Use this workflow to debug unexpected tool movements.

5.3 Verifying Feed Rates

1. Load the G-code program.
2. Click **Play** to start the simulation.
3. Watch the F value in the Current State panel.
4. Review the Feed Rate Range in the statistics panel.
5. Verify that feed rates are within your machine's capabilities.

5.4 Checking Bounding Box

1. Load and parse the G-code program.
2. Review the Bounding Box statistics (X, Y, Z extents).
3. Verify that the toolpath fits within your machine's work envelope.
4. Check for unexpected negative coordinates or excessive travel.

6 Architecture

This section presents the UML diagrams for the G-Code Viewer application.

6.1 Architecture Overview

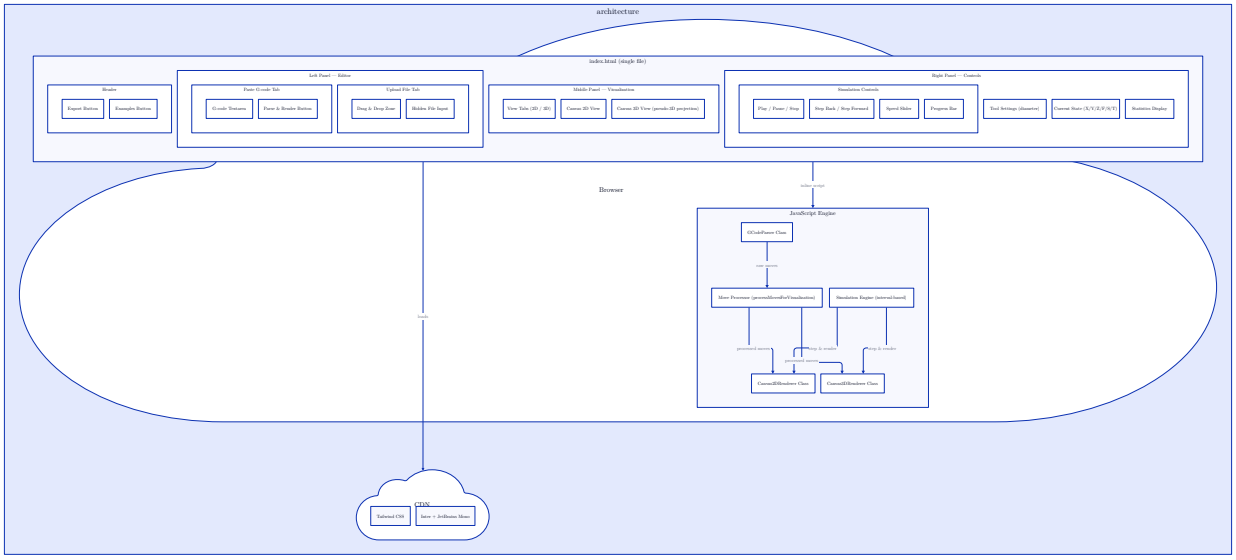

Figure 1: System architecture showing the parser, simulation engine, and rendering components.

6.2 Sequence Diagram — Main Parse and Render Flow

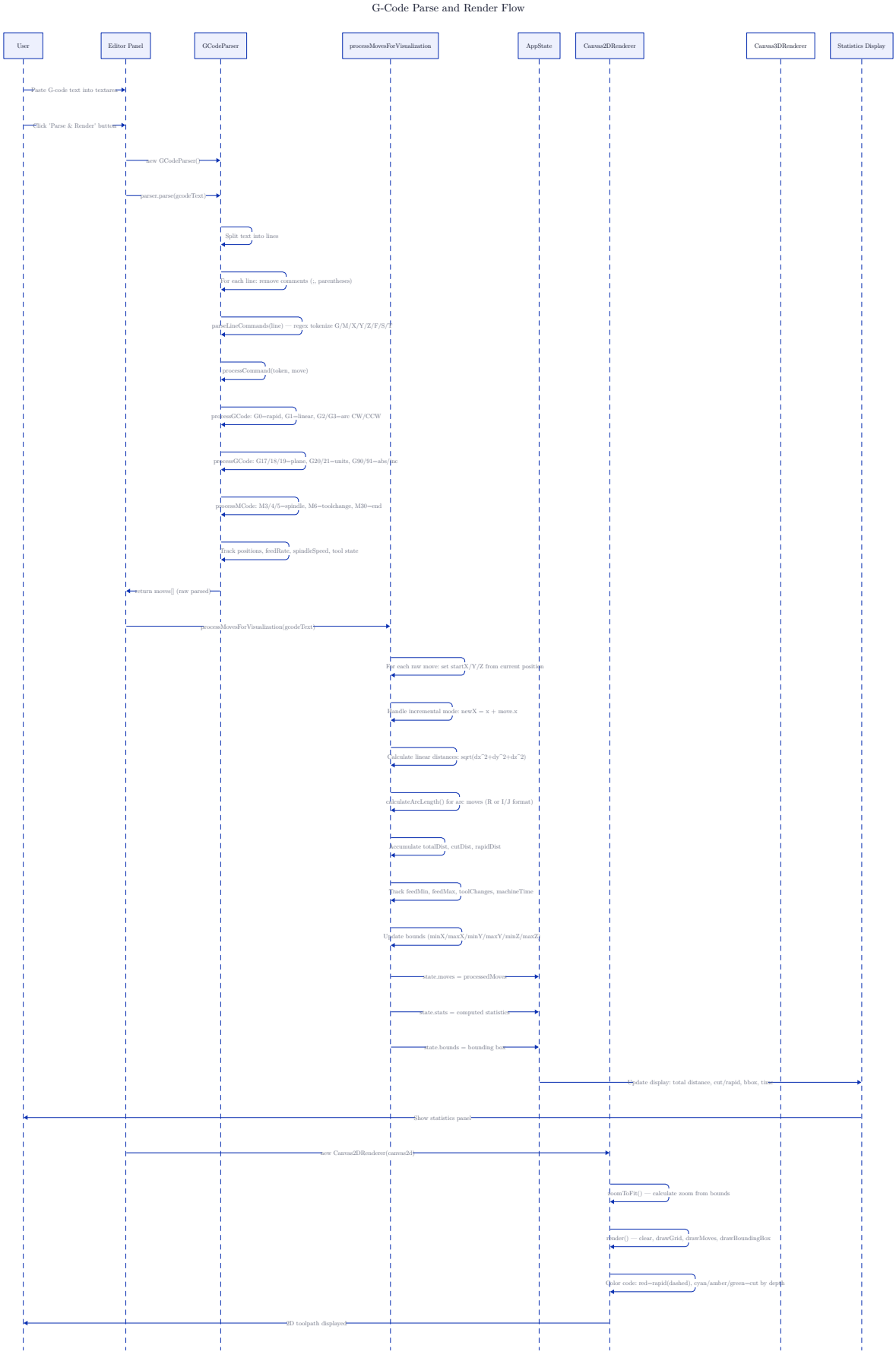

Figure 2: Sequence diagram showing the G-code parsing, move generation, and canvas rendering flow.

6.3 Sequence Diagram — File Upload

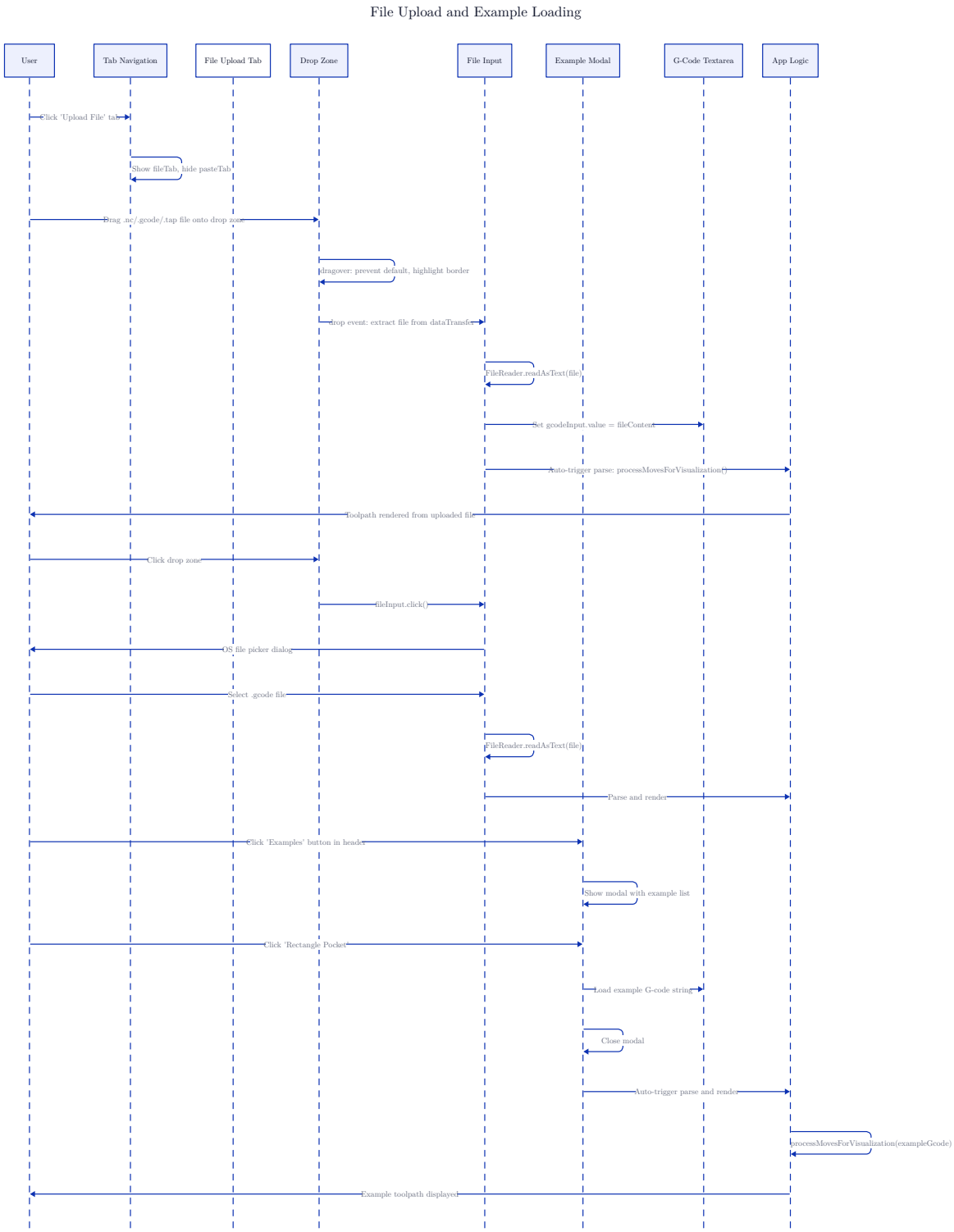

Figure 3: Sequence diagram for the file upload and drag-and-drop workflow.

6.4 Sequence Diagram — Simulation

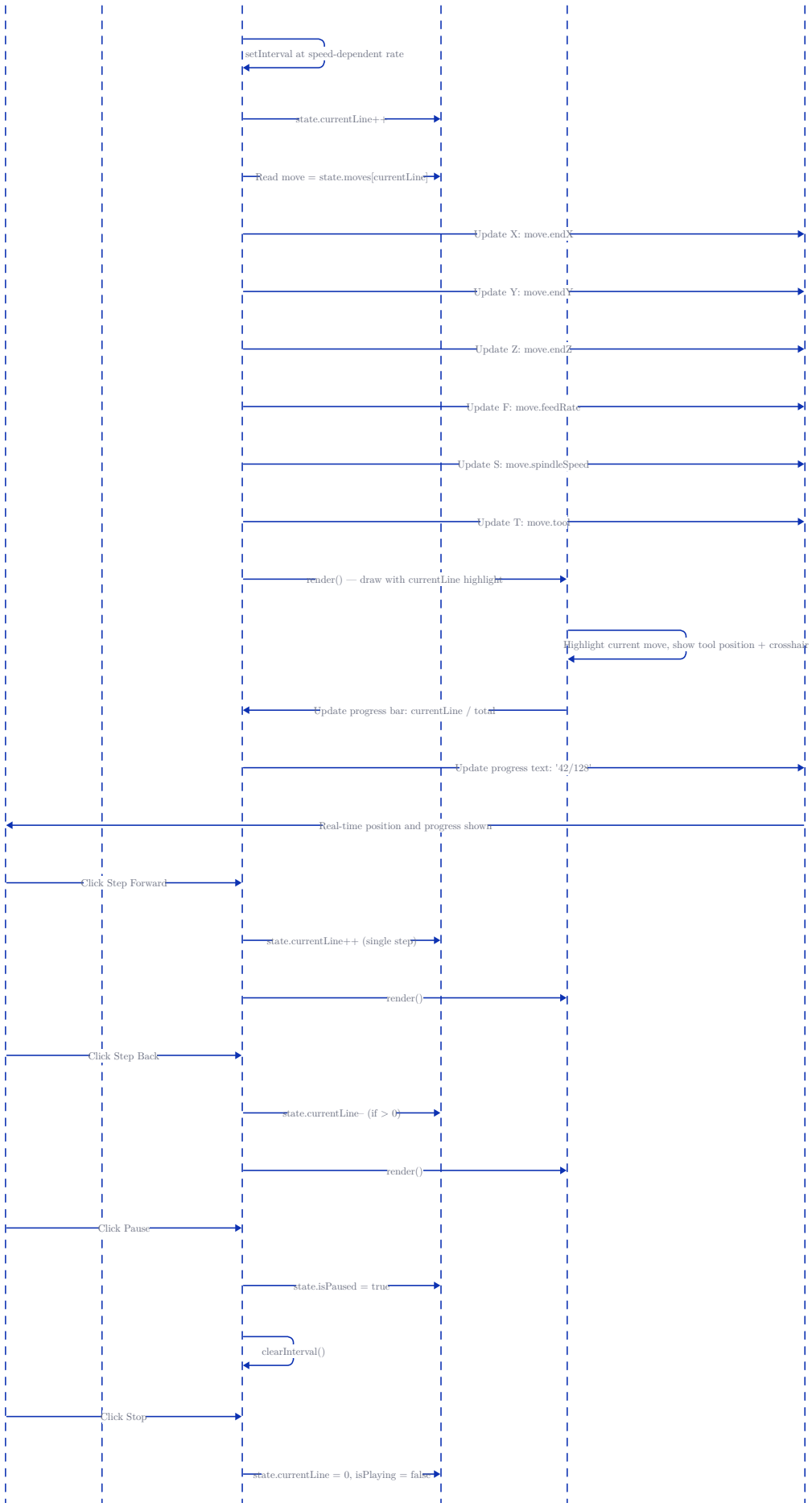

Figure 4: Sequence diagram showing the simulation playback loop with state updates.

6.5 State Diagram

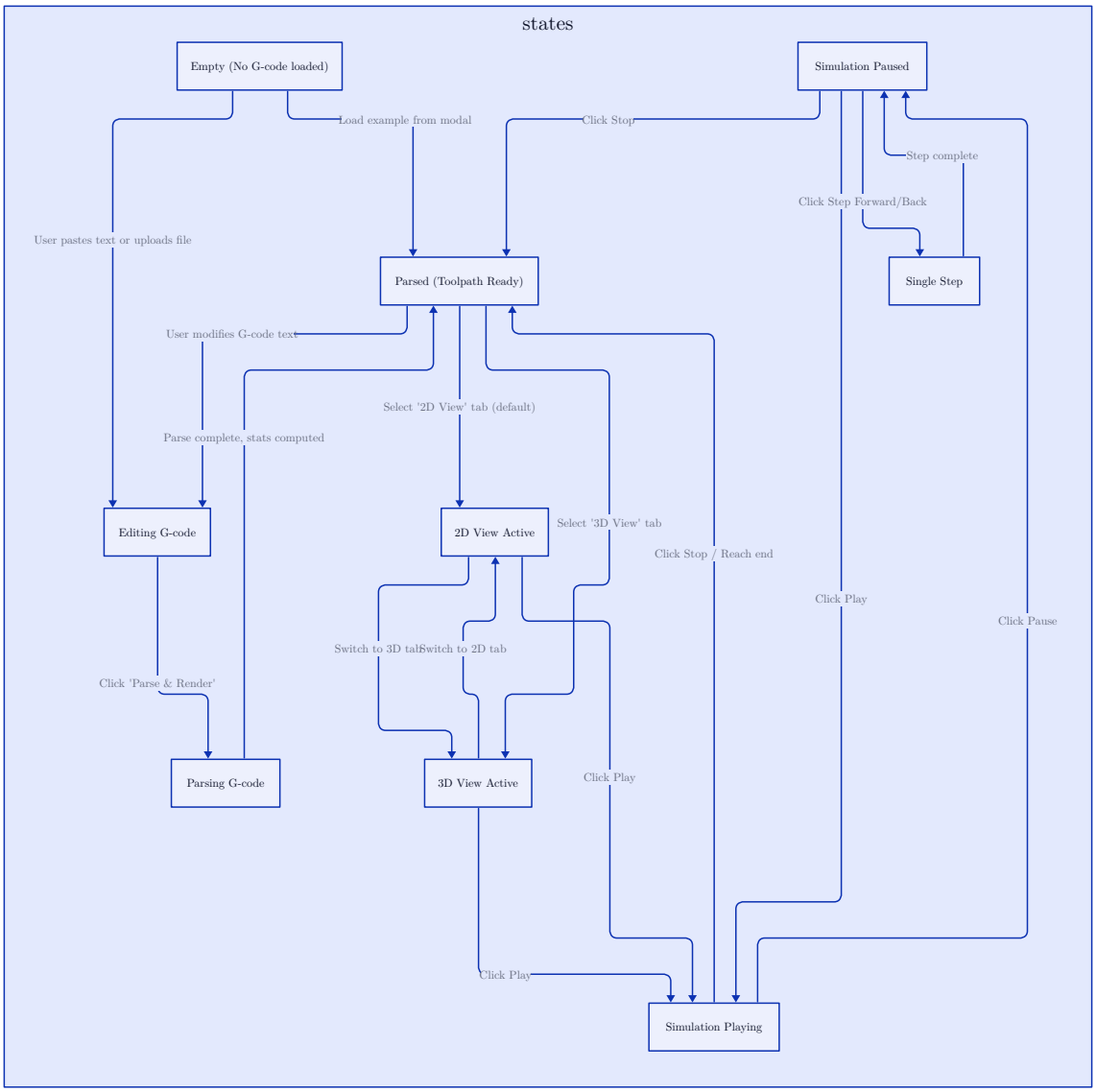

Figure 5: State diagram showing parser, simulation, and view mode states.

7 Configuration

7.1 Supported File Extensions

The file upload accepts the following extensions:

Extension	Description
`.nc`	Numerical control (standard)
`.gcode`	G-code (common for 3D printers)
`.tap`	CNC tap file
`.ngc`	LinuxCNC native format

`.txt`	Plain text G-code

7.2 Default Tool Settings

- **Tool Diameter**: 3.175 mm (1/8 inch end mill)
- **Units**: Millimeters (G21 default, switchable via G20)
- **Feed Mode**: Absolute (G90 default)
- **Plane**: XY (G17 default)

8 Troubleshooting

8.1 G-Code Not Parsing

- Ensure G-code commands follow standard syntax (e.g., `G1 X10 Y20 F500`).
- Comments must use `;` or parentheses `( )` notation.
- Check for unsupported or proprietary machine codes.
- The parser handles the most common G/M codes; machine-specific extensions may not be recognized.

8.2 3D View Not Rendering

- Verify that WebGL is enabled in your browser.
- Some older or integrated graphics drivers may not support WebGL. Try updating drivers.
- Check the browser console for rendering errors.

8.3 Simulation Stuttering

- Reduce the simulation speed.
- Large G-code programs (10,000+ lines) may cause slower rendering. Allow time for the initial parse.
- Close other browser tabs to free resources.

8.4 Incorrect Bounding Box

- Verify the correct units mode (G20 for inches, G21 for mm).
- Check for G92 coordinate offsets that may shift the apparent toolpath.
- Arc interpolation (G2/G3) may extend beyond the start/end points.

8.5 File Upload Not Working

- Ensure the file has a supported extension (.nc, .gcode, .tap, .ngc, .txt).
- Check that the file is not empty.
- Try pasting the file contents directly into the Paste tab as a workaround.